



Government Welfare & Social Security

AI-Based NSAP Scheme Eligibility Prediction using IBM Cloud and Machine Learning



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THE CHALLENGE

Problem Statement

The National Social Assistance Programme (NSAP) is a flagship welfare initiative of the Government of India that provides financial support to elderly citizens, widows, and persons with disabilities belonging to Below Poverty Line (BPL) households.

Each welfare scheme has different eligibility criteria, making manual verification difficult, time-consuming, and prone to human error.

This project develops an AI-based multi-class machine learning model that predicts the most appropriate NSAP welfare scheme using demographic and socio-economic information, helping government agencies automate applicant classification and ensure deserving beneficiaries receive assistance efficiently.



Who is affected

Elderly citizens, widows &
persons with disabilities in
BPL households

Core issue

Manual, slow & error-prone
eligibility verification

Proposed Solution

An AI-powered NSAP Scheme Eligibility Prediction System built using IBM Cloud and Machine Learning.

01



Collect & Preprocess

Gather applicant demographic and socio-economic data

02



Feature Engineering

Normalize data and craft predictive features

03



Train Models

Train multiple ML classification algorithms

04



Select Best Model

Choose the top performer using evaluation metrics

05



Deploy on IBM Cloud

Serve the trained model via a Flask API

06



Real-Time Prediction

Predict the suitable NSAP scheme instantly

Technology Used



IBM Cloud

Secure infrastructure for model development and deployment



Python

Core language for data preprocessing and model development



Pandas & NumPy

Data cleaning and preprocessing



Scikit-Learn

Machine learning model training and evaluation



Random Forest

Best performing classification algorithm



Flask

REST API for real-time prediction

MODEL PERFORMANCE

Random Forest Classifier

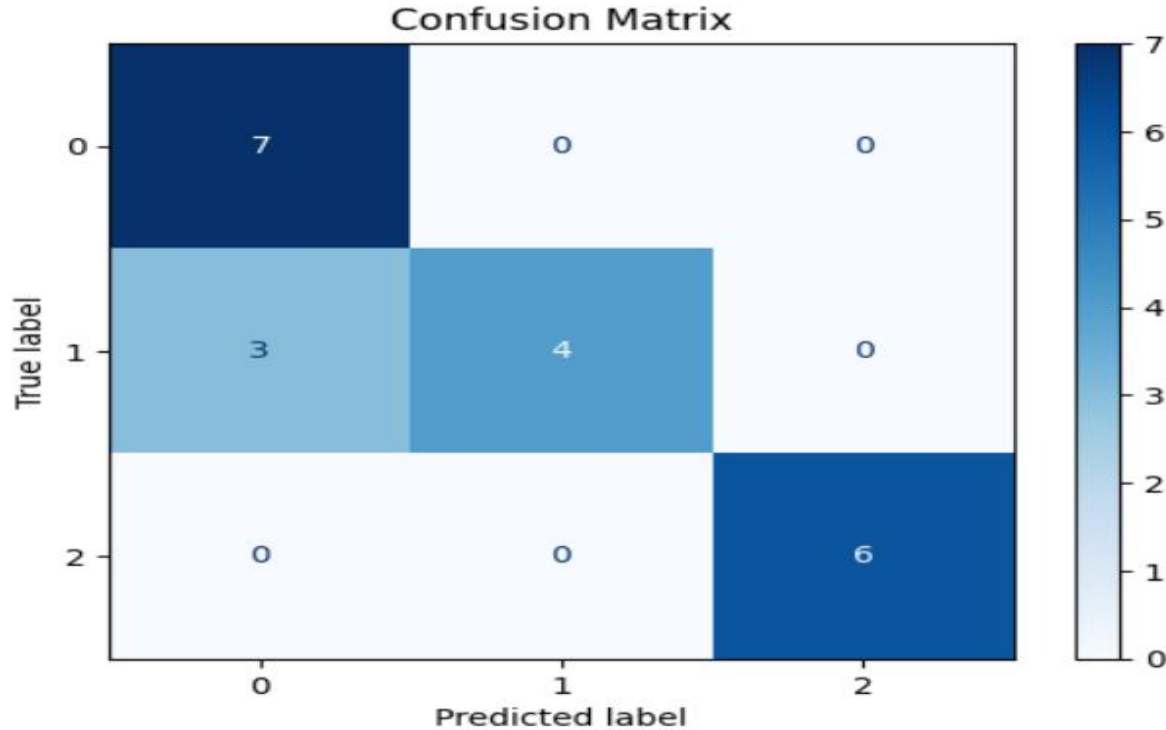
85%

Accuracy Obtained

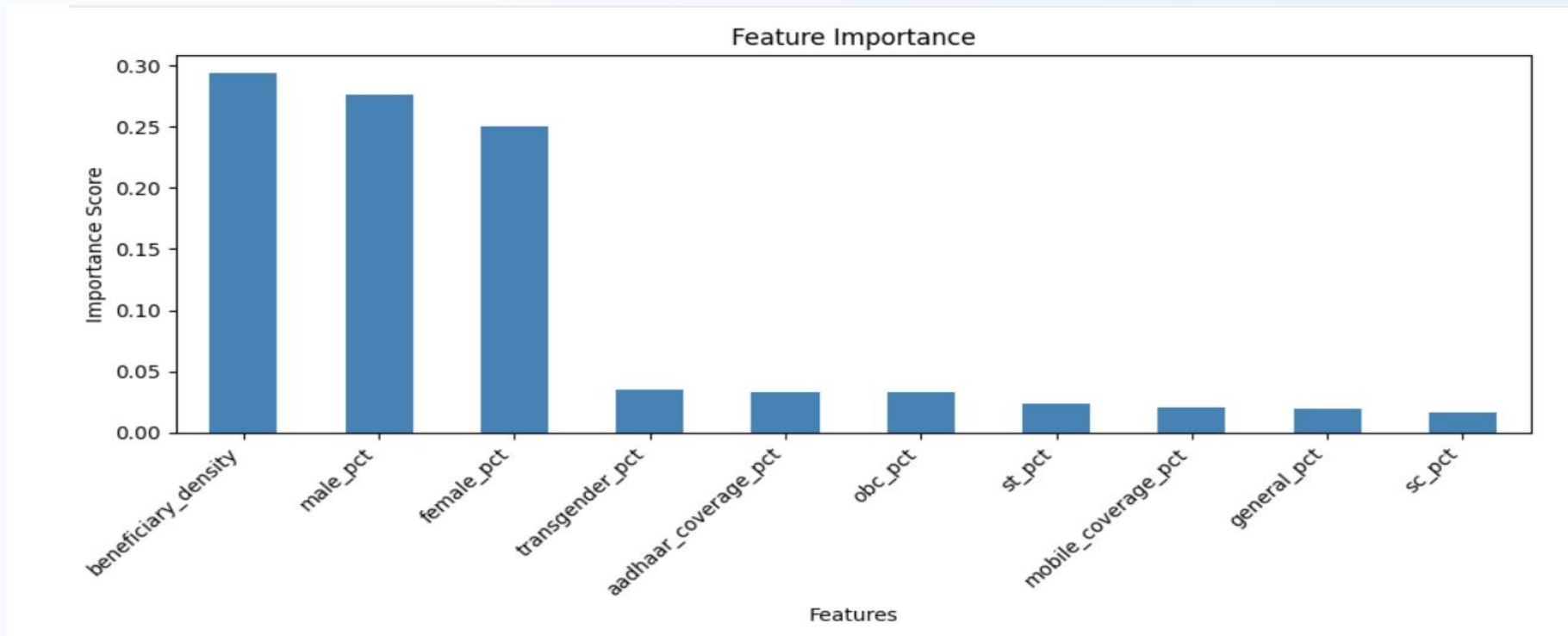
HYPERPARAMETER TUNING

- Number of Trees (Estimators)
- Maximum Tree Depth
- Random State Optimization
- Train-Test Split

Training Screenshot



Best Model Screenshot



Testing Screenshot

:

	Actual Scheme	Predicted Scheme
0	IGNDPS	IGNDPS
1	IGNWPS	IGNWPS
2	IGNDPS	IGNDPS
3	IGNDPS	IGNDPS
4	IGNOAPS	IGNDPS

Prediction Output Screenshot

WHAT SETS IT APART

Novelty and Uniqueness



Automates NSAP scheme eligibility prediction using Artificial Intelligence



Reduces manual verification time and administrative workload



Provides fast and accurate multi-class classification



Improves transparency and consistency in welfare distribution



Uses feature engineering to improve prediction quality



Easily deployable on IBM Cloud for real-world government applications



Scalable architecture capable of supporting additional welfare schemes

SOURCE CODE

GitHub Repository



REPOSITORY NAME

**NSAP-Scheme-
Eligibility-Prediction**

GitHub repository link :-

<https://github.com/munniramsaraswat33/NSAP-Scheme-Eligibility-Prediction.git>

REPOSITORY SHOULD CONTAIN

- | | |
|----------------|-------------------------|
| ✓ README.md | ✓ requirements.txt |
| ✓ Dataset | ✓ Dockerfile |
| ✓ IBM Notebook | ✓ Presentation |
| ✓ app.py | ✓ Problem Statement PDF |

Future Scope

01



Integration with Aadhaar and Digital India platforms

02



Real-time applicant verification

03



Support for additional government welfare schemes

04



Improved prediction using Deep Learning

05



Mobile application for citizens

06



Dashboard for government officials

07



Explainable AI for transparent decision-making

Credly Certificate



Credly Certificate



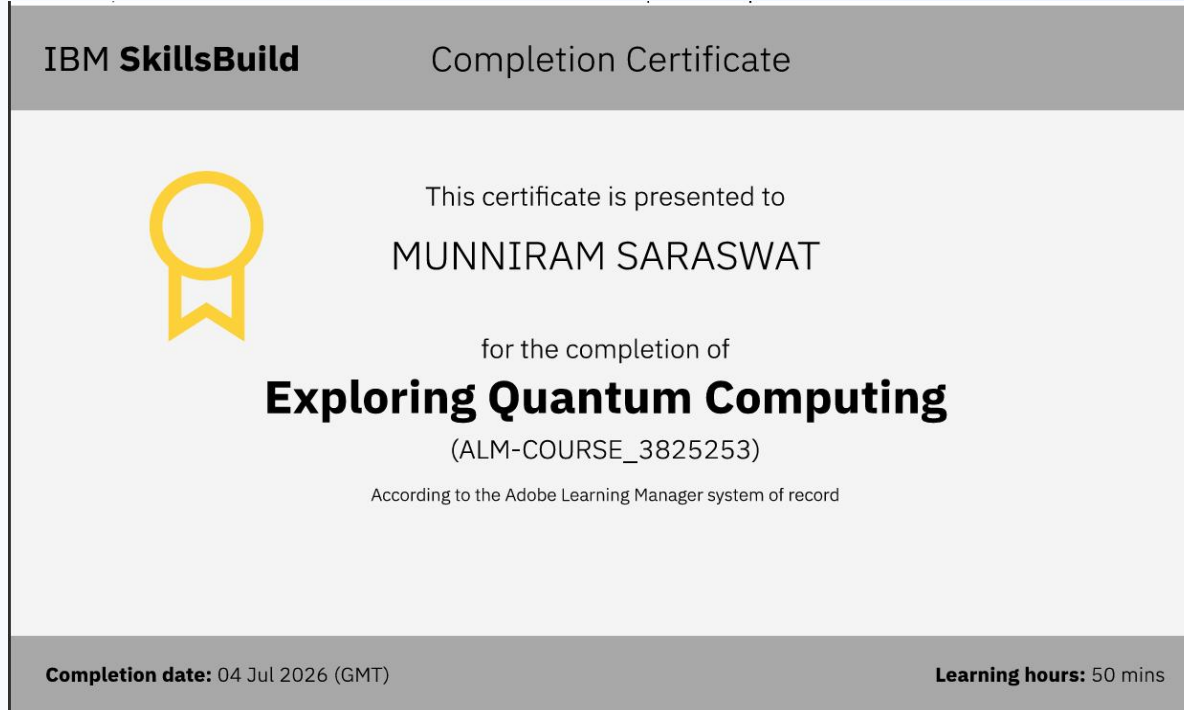
Credly Certificate



Completion Certificate



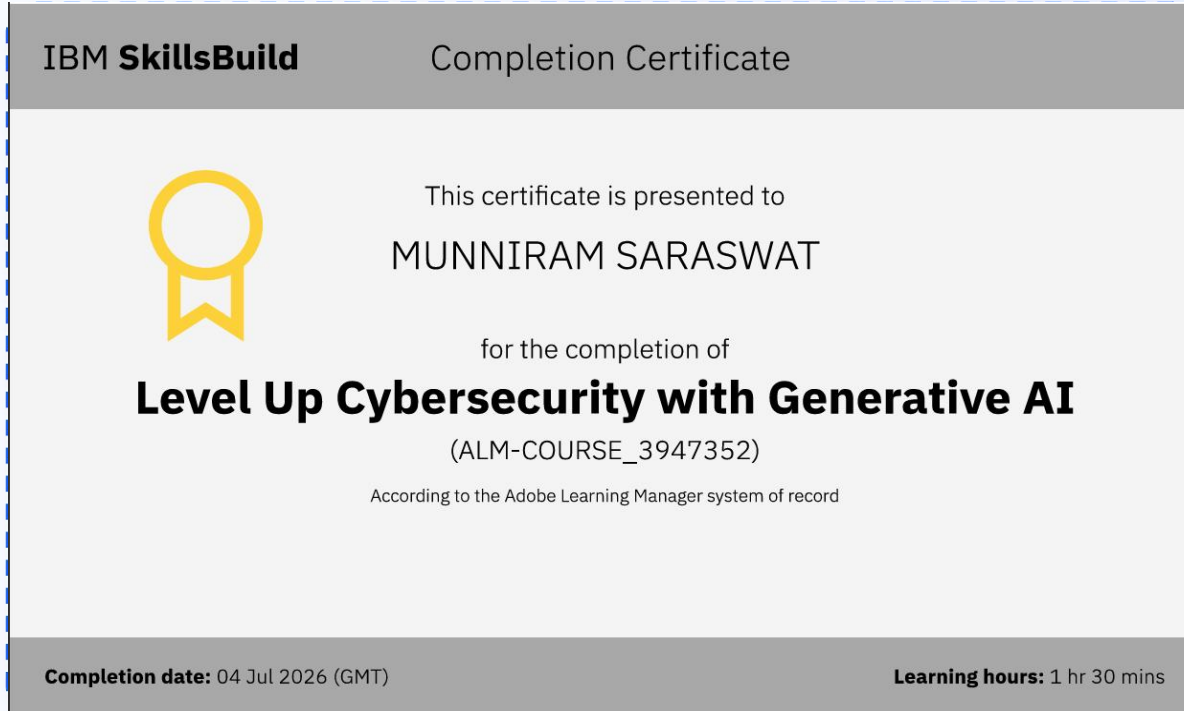
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Completion Certificate

